

Education

- 2012–2017 PhD in Computer Science, IMG Lab
University of Saskatchewan
- 2009–2012 MEng in Mechanical Engineering and Automatic Control
Northwestern Polytechnical University
- 2005–2009 BEng in Mechanical Engineering
Hubei University of Technology

Areas of Expertise

Medical Image Analysis, Machine Learning, Deep Learning, Image Segmentation

Experience

- 2023–present Senior AI Lead, University of Saskatchewan, Saskatoon, Canada
- Leading translational medical imaging AI research in collaboration with clinical and academic investigators.
 - Designing and developing machine learning methodologies for clinically relevant imaging applications, including cardiovascular imaging and diagnostic decision support.
 - Co-authoring manuscripts, conference presentations, and technical reports for peer-reviewed dissemination.
 - Developing competitive grant applications from project conception through technical writing and study design.
 - Co-leading a multidisciplinary research team, contributing to personnel recruitment, trainee supervision, and mentorship of graduate students and junior researchers.
 - Contributing to research infrastructure development, computational workflows, and reproducible AI experimentation pipelines.
 - Secured two CoMRAD research grants as Co-PI (CAD \$30,000 each) supporting translational medical imaging AI research, and obtained a Digital Research Alliance of Canada RAC computational resource allocation valued at approximately CAD \$18,000.
- 2023–present Staff Research Scientist, Circle Cardiovascular Imaging, Calgary, Canada
- Led the development of an FDA-cleared coronary plaque analysis AI algorithm for 3D cardiac CT imaging.
 - Investigating and integrating emerging machine learning technologies and architectures to advance next-generation cardiovascular imaging AI solutions.
 - Leading the evolution of the machine learning development and deployment stack, including scalability, infrastructure integration, and enterprise-level deployment workflows.
 - Collaborating with multidisciplinary teams across research, engineering, and clinical domains to translate AI research into production-ready clinical applications.
- 2021–2023 Senior Software Research Engineer, Circle Cardiovascular Imaging, Calgary, Canada
- Led improvements to the internal machine learning development infrastructure, including dataset versioning, experiment tracking, and standardization of development workflows and coding practices.
 - Led the migration and upgrade of TensorFlow within the internal machine learning inference engine to support scalability, maintainability, and deployment of advanced AI models.
 - Contributed to the design and optimization of reproducible machine learning pipelines for clinical imaging AI development.

- 2019–2021 Software Research Engineer, Circle Cardiovascular Imaging, Calgary, Canada
- Led the development of automated left atrial appendage (LAA) segmentation methods for interventional planning using 3D cardiac CT.
 - Led the development of automated left atrial (LA) wall segmentation methods for pre-procedural planning using 3D cardiac CT.
 - Led the development of coronary centerline extraction and branch labeling algorithms for coronary artery analysis on 3D cardiac CT.
 - Led the regulatory documentation and technical validation processes that contributed to FDA clearance of the coronary analysis product.
 - Developed automated chamber segmentation methods for cardiac function analysis on cardiac MRI.
 - Developed automated landing zone detection methods for Watchman device planning on 3D cardiac CT.
- 2017–2019 Postdoctoral Fellow, University of Saskatchewan, Saskatoon, Canada
- Developed deep learning methods for low-dose CT denoising using generative adversarial networks.
 - Investigated unsupervised and semi-supervised adversarial learning approaches for melanoma recognition.
 - Developed automated catheter and tube detection methods for pediatric chest radiographs.
 - Developed ensemble-learning approaches for pneumonia detection on adult chest X-ray images.
 - Conducted comprehensive literature reviews on adversarial learning techniques in medical image analysis.
 - Conducted a comprehensive review of computer-aided assessment methods for catheter and tube placement evaluation.
- 2016–2017 Research Assistant, University of Saskatchewan, Saskatoon, Canada
- Provided development and technical support for the College of Education internship management platform [Placement](#).
 - Contributed to web application development, server maintenance, and technical infrastructure support.
- 2012–2016 Research Assistant, University of Saskatchewan, Saskatoon, Canada
- Coordinated a laboratory research journal club, organizing weekly paper selections and facilitating scientific discussion on emerging topics in computer vision and medical image analysis.
 - Developed computer vision and image analysis methods for automated weed seed species recognition in collaboration with the Canadian Food Inspection Agency (CFIA).
 - Developed image processing methods for single-image defocus blur segmentation and analysis.
 - Contributed to research software development, experimental design, and quantitative evaluation of image analysis algorithms.
- 2014 Teaching Assistant, University of Saskatchewan, Saskatoon, Canada
- CMPT 280 - Intermediate Data Structures and Algorithms
- 2012, 13 Teaching Assistant, University of Saskatchewan, Saskatoon, Canada
- CMPT 111 - Introduction to Computer Science and Programming
 - CMPT 115 - Principles of Computer Science
- 2012 Software Development Engineer, Zhongxing Telecommunication Equipment Corporation (ZTE), Xi'an, China

Awards and Achievements

- 2021 VIP Innovation Champion (Circle Cardiovascular Imaging)
- 2021 Rank 46 out of 1305 teams (solo, top 4%) in [SIIM-FISABIO-RSNA COVID-19 Detection](#)
- 2019 Certificate of Merit (RSNA annual meeting)
- 2019 2nd place in the Scientific Research Project (CAR annual meeting)
- 2018 Rank 14 out of 1499 teams (solo, top 1%) in [RSNA Pneumonia Detection Challenge](#)
- 2015 Best Poster Award (ISTA annual meeting)
- 2014 Student Travel Award

- 2010 Excellent Student Cadre of Marine College
- 2009 Outstanding Graduate of Hubei University of Technology
- 2006, 07, 08 Elite Student Scholarship
- 2007 National Scholarship (top 3% students), by Ministry of Education of the People's Republic of China

Skills

- Programming Skills** Matlab, Python (*PyTorch*, *TensorFlow*), C/C++ (*OpenCV*, *VLFeat*, *Qt5*), HTML/CSS/JavaScript (*Flask*, *Django*), Lua (*Torch*), SQL, $\text{\LaTeX}$
- Languages** Chinese Mandarin (*Native*), English (*Fluent*)

Professional Activities

- Conference Reviewer for CRV 2019, MICCAI 2019/2021/2022/2023, MIDL 2020
- Journal Reviewer for MedIA, TMI, TIP, JBHI, SPL, etc

Publications

- [1] Pouya Shiri, Xin Yi, Neel P Mistry, Samaneh Javadinia, Mohammad Chegini, Seok-Bum Ko, Amirali Baniasadi, and Scott J Adams. Generating synthetic contrast-enhanced chest ct images from non-contrast scans using slice-consistent brownian bridge diffusion network. In *Proceedings of STACOM Workshop at MICCAI 2025*, 2025.
- [2] Lisette Lockhart, Xin Yi, Nathan Cassady, Alexandra Nunn, Cory Swingen, and Alborz Amir-Khalili. Automatic landing zone plane detection in contrast-enhanced cardiac ct volumes. In *International Workshop on Statistical Atlases and Computational Models of the Heart*, pages 240–249. Springer, 2023.
- [3] Maurice Pradella, Michael B Scott, Muhammad Omer, Seth K Hill, Lisette Lockhart, Xin Yi, Alborz Amir-Khalili, Alireza Sojoudi, Bradley D Allen, Ryan Avery, et al. Fully-automated deep learning-based flow quantification of 2d cine phase contrast mri. *European Radiology*, 33(3):1707–1718, 2023.
- [4] Robert DE Henderson, Sirwa Padash, Scott J Adams, Carolyn Augusta, Xin Yi, and Paul Babyn. Neonatal catheter and tube placement and radiographic assessment statistics in relation to important anatomic landmarks. *American Journal of Perinatology*, 2023.
- [5] Scott J Adams, Robert DE Henderson, Xin Yi, and Paul Babyn. Artificial intelligence solutions for analysis of x-ray images. *Canadian Association of Radiologists Journal*, 72(1):60–72, 2021.
- [6] Robert DE Henderson, Xin Yi, Scott J Adams, and Paul Babyn. Automatic detection and classification of multiple catheters in neonatal radiographs with deep learning. *Journal of Digital Imaging*, 34(4):888–897, 2021.
- [7] Xin Yi, Scott J Adams, Robert DE Henderson, and Paul Babyn. Computer-aided assessment of catheters and tubes on radiographs: How good is artificial intelligence for assessment? *Radiology. Artificial intelligence*, 2(1), 2020.
- [8] Xin Yi, Ekta Walia, and Paul Babyn. Generative adversarial network in medical imaging: A review. *Medical image analysis*, page 101552, 2019.
- [9] Xin Yi, Scott Adams, Paul Babyn, and Abdul Elnajmi. Automatic catheter and tube detection in pediatric x-ray images using a scale-recurrent network and synthetic data. *Journal of digital imaging*, pages 1–10, 2019.

- [10] Xin Yi and Paul Babyn. Sharpness-aware low-dose ct denoising using conditional generative adversarial network. *Journal of Digital Imaging*, Feb 2018.
- [11] Xin Yi, Walia Ekta, and Paul Babyn. Unsupervised and semi-supervised learning with categorical generative adversarial networks assisted by wasserstein distance for dermoscopy image classification. *arXiv*, 2018.
- [12] Xin Yi and Mark Eramian. Lbp-based segmentation of defocus blur. *IEEE Transactions on Image Processing*, 25(4):1626–1638, 2016.
- [13] Xin Yi, Mark Eramian, Ruoqing Wang, and Eric Neufeld. Identification of morphologically similar seeds using multi-kernel learning. In *Computer and Robot Vision (CRV), 2014 Canadian Conference on*, pages 143–150. IEEE, 2014.

Conference Presentations

- 2025 Radiology Society of North America annual meeting (RSNA), Chicago, Illinois (educational exhibit). *Diffusion Without Infusion: A Review of Diffusion Models for Synthetic Contrast-Enhanced Imaging*.
- 2019 Radiology Society of North America annual meeting (RSNA), Chicago, Illinois (educational exhibit). *Computer-Aided Assessment of Catheters and Tubes on Radiographs: How Good is Artificial Intelligence for Assessment?*.
- 2019 Canadian Association of Radiologists annual meeting (CAR), Montréal, QC (oral). *Deep learning for automatic multi-catheter detection on pediatric radiographs*.
- 2018 International conference on Medical Imaging with Deep Learning (MIDL), Amsterdam, Netherland (poster). *Automatic catheter detection in pediatric X-ray images using a scale-recurrent network and synthetic data*.
- 2018 Canadian Association of Radiologists annual meeting (CAR), Montréal, QC (poster). *Evaluation of low-dose CT denoising on patient data using sharpness-aware conditional generative adversarial network*.
- 2017 The Scientific Conference on Machine Intelligence in Medical Imaging (C-MIMI) of the Society for Imaging Informatics in Medicine (SIIM), Baltimore, MD (oral). *Sharpness-aware Low dose CT denoising using conditional generative adversarial network*.
- 2017 The Scientific Conference on Machine Intelligence in Medical Imaging (C-MIMI) of the Society for Imaging Informatics in Medicine (SIIM), Baltimore, MD (oral). *Unsupervised and Semi-supervised learning with Categorical Generative Adversarial Networks for Dermoscopic Image Classification*.
- 2016 The International Seed Testing Association annual meeting, Tallinn, Estonia (poster). *Preliminary validation study using computer vision for seed identification*.
- 2015 The International Seed Testing Association annual meeting, Montevideo, Uruguay (poster, **first prize**). *Seed identification using computer vision techniques*.
- 2014 Conference on Computer and Robot Vision, Montréal, QC (oral). *Identification of morphologically similar seeds using multi-kernel learning*.